

Motivation

Landslides cause **severe casualties and infrastructure damage** across Sri Lanka's central highlands. Post-event mapping is critical for emergency response and risk reduction, yet conventional visual interpretation of drone data is time-consuming and operator-dependent especially when multiple simultaneous events occur after a single rainfall episode.

Deep learning segmentation offers a path toward semi-automated mapping. Its performance, however, depends critically on which input data channels are available. This study asks:

"Does fusing LiDAR-derived terrain features with high resolution RGB drone imagery improve U-Net landslide segmentation compared to spectral data alone?"

Study Area & Dataset

12 post-landslide drone survey sites across Sri Lanka's central highlands:



Figure 1: Overview of the 12 drone landslide survey sites in Sri Lanka

Input data:

- LiDAR point cloud → ground-classified DTM at 0.1 m grid
- RGB orthomosaic at < 5 cm GSD
- Derived layers: Hillshade (azimuth 315°, elevation 45°) and Slope

Coordinate reference system: EPSG:5235 (Sri Lanka Datum 1999)

Train / Validation / Test split:

Sites 01–10 → 80%: Training 20% Validation
Sites 11 & 12 → hold for test

Tiling & class balance:

- 256 × 256 px tiles, stride-based sampling
- Sites 1,3,5,6,8,9: fully tiled; sites 2,4,7,10: partially tiled
- Landslide pixels ≈ 2% of each scene

Labelling:

Binary masks (landslide / background) manually delineated per site from orthomosaic and DSM visual inspection

Input Configurations

- RGB**
3 channels: red, green, blue from orthomosaic
- RGB + DTM**
4 channels: + digital terrain model at 0.1 m
- RGB + DTM + Hillshade**
5 channels: + terrain illumination
- RGB + DTM + Hillshade + Slope**
6 channels: + slope gradient

U-Net Architecture & Training

Deep Learning Architecture

Model: U-Net based semantic segmentation framework

Architecture

Encoder–decoder network with **4 downsampling levels** and **skip connections**

Encoder:

3×3 Convolution → Batch Normalization → ReLU activation
2×2 Max pooling (4 stages)

Bottleneck:

Deep feature representation with **1024 channels**

Decoder:

Bilinear upsampling
Feature fusion through encoder skip connections at each scale

Output Layer:

1×1 Convolution → Sigmoid activation
Generates **binary landslide probability map**

Training hyperparameters:

Parameter	Configuration
Learning Rate	1×10^{-4}
Batch Size	8
Training Epochs	25
Loss Function	Binary Cross-Entropy with class weighting
Input Tile Size	256×256 pixels

Terrain feature derivation:

- DTM: ground-classified LiDAR rasterised at 0.1 m
- Hillshade: sun azimuth 315°, elevation 45° (single illumination)
- Slope: first-order gradient magnitude from DTM (degrees)

Inference & evaluation:

- Sigmoid threshold: 0.5 (default; not optimised)
- Metrics: Precision, Recall, F1-Score at pixel level
- Accuracy reported but not used for model selection (misleading under class imbalance)

Results

Configuration	Acc.	Prec.	Recall	F1
RGB	0.978	n/a	≈ 0	n/a
RGB + DTM	0.978	0.482	≈ 0	≈ 0
RGB + DTM + Hillshade	0.978	n/a	≈ 0	n/a
RGB + DTM + Hillshade + Slope	0.977	0.491	0.736	0.589
└ Site 11 (Uthuwana Kanda)	0.974	0.474	0.841	0.606
└ Site 12 (Victoria Reservoir)	0.982	0.536	0.568	0.552

Table 1: Overall metrics

Performance Comparison

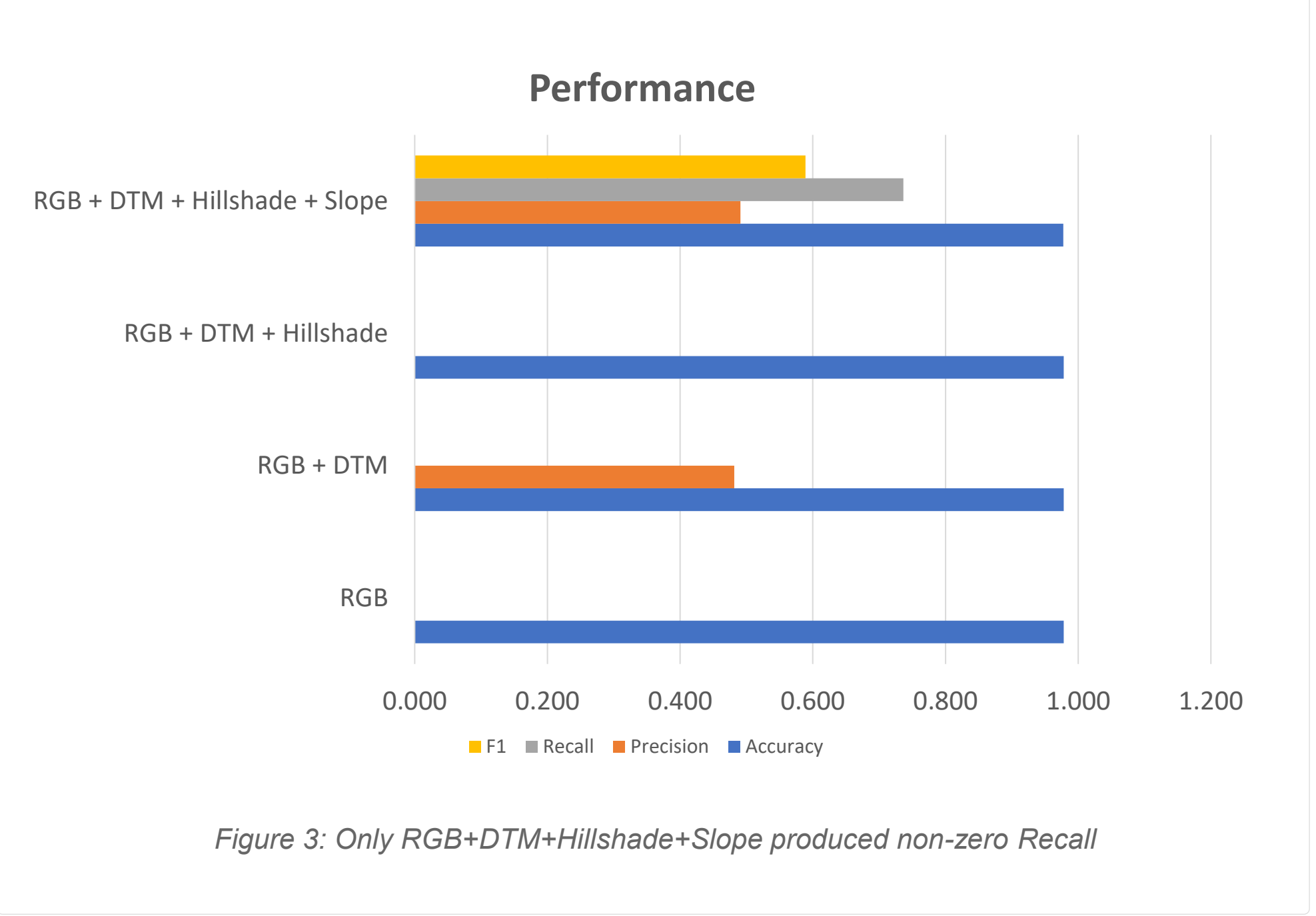


Figure 3: Only RGB+DTM+Hillshade+Slope produced non-zero Recall

Discussion

Only RGB + DTM + Hillshade + Slope produced a functional detector (F1 = 0.589, Recall = 0.736), correctly flagging the majority of landslide pixels on both held out sites. All three simpler configurations collapsed to predicting background everywhere.

Accuracy is misleading due to severe class imbalance: landslide pixels account for less than 2% of each scene. All four models score about 97.7% to 97.8% accuracy, including those that detect nothing. F1 and Recall are the decisive metrics.

Key limitation: inverted pos_weight. The loss used pos_weight = 0.2, which reduces the weight of the minority landslide class instead of increasing it. This is the most likely reason 3 of 4 models converged to the trivial solution of predicting background everywhere. Correcting pos_weight to about 5 to 10 is the most critical fix before treating these results as a definitive ablation study.

Terrain shape matters: the requirement for slope and hillshade is consistent with the intuition that landslide scars exhibit characteristic terrain discontinuities such as steep, freshly exposed soil with sharp slope breaks, which spectral colour alone cannot reliably distinguish from other bare surfaces.

Sparse tile sampling due to dataset scale: source orthomosaics are very large (up to about 58,600 by 45,600 pixels per site), so tiles were sampled every 768 pixels rather than exhaustively during both training data generation and prediction, to keep processing feasible. Dense tiling of all 10 sites was estimated to take over 20 hours. This means training coverage is uneven across sites (one site fully tiled, others only partially), and prediction rasters have gaps between sampled tiles rather than full wall to wall coverage.

Training & Validation Loss Curves

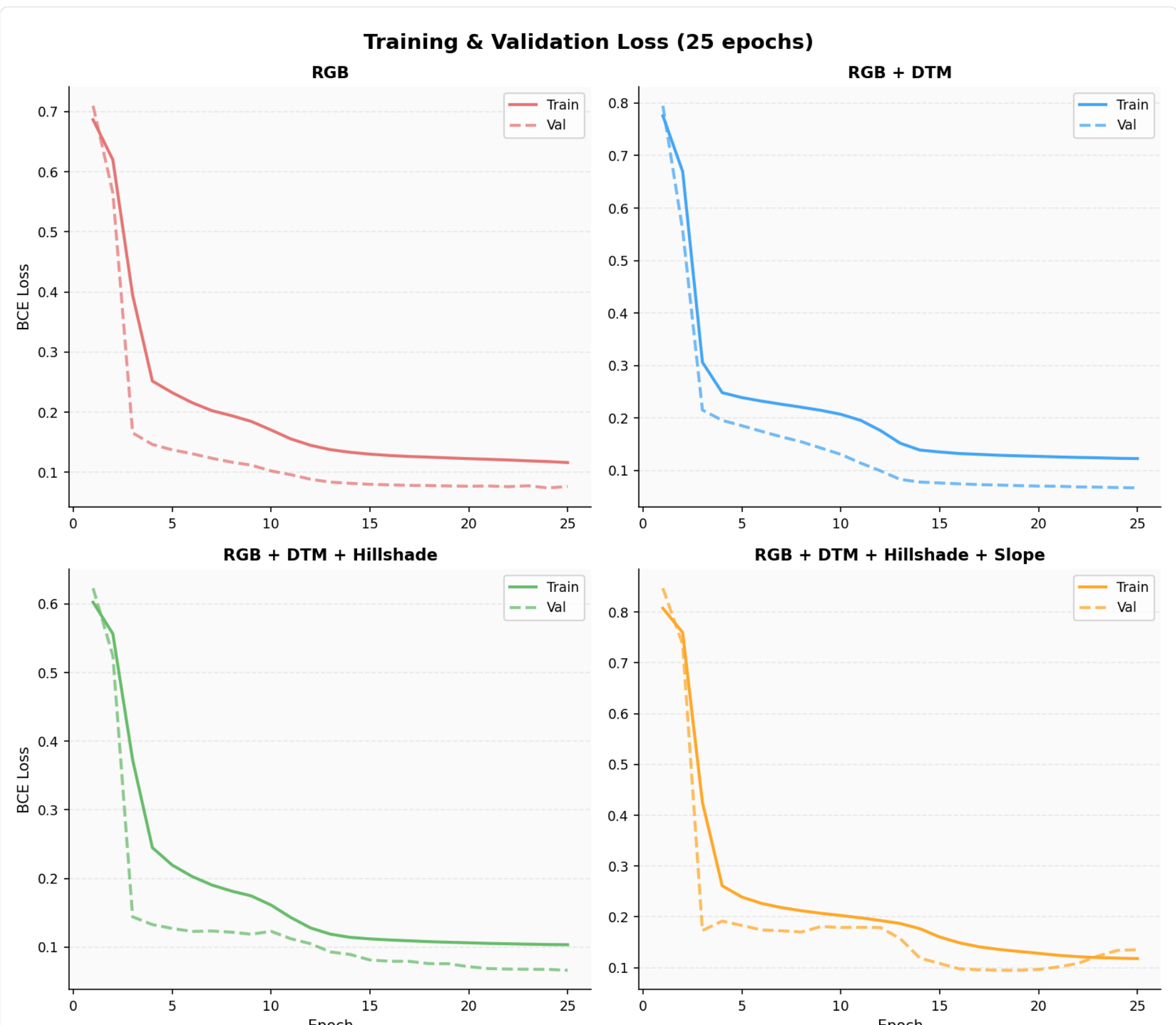


Figure 2: Train / validation loss for all four configurations (25 epochs)